Traditionally, a conic section is created by intersecting a conic surface with a plane. Through these means a circle, ellipse, hyperbola, parabola, line, two lines, or point may be created. What sections can be created by intersecting a conic surface with *another* conic surface? How many permutations of placement are possible?

For simplicity's sake, I will be using right cones, and the two intersecting conic surfaces will be identical in angle at the vertex.

I will explore intersection through translation. Since neither conic surface will be rotating, I can use the position of the vertex as the sole indicator of the overall position of the conic surface. There are four basic placements of a vertex relative to the other cone.

1. Both vertices share the same space.
2. The moved vertex lies within the bounds of the other cone. (i.e. above or below, but still within the angle at the vertex)
3. The moved vertex lies outside of the other cone
4. The moved vertex lies on a generatrix of the other cone.

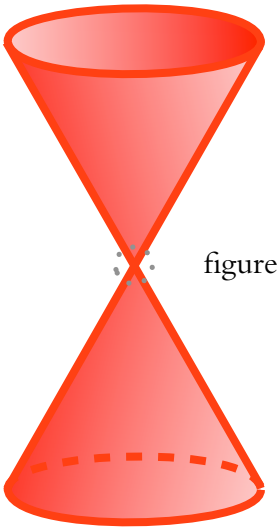
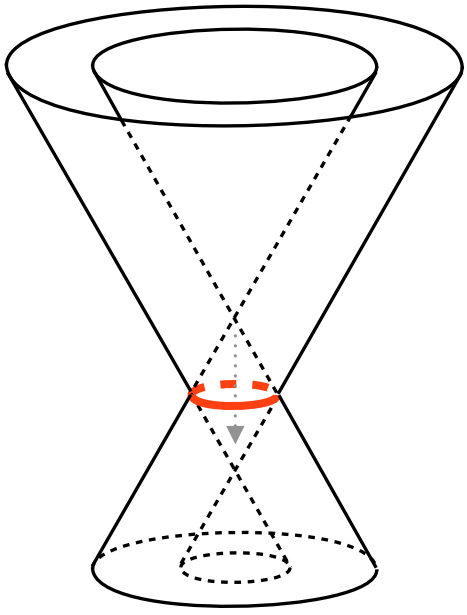


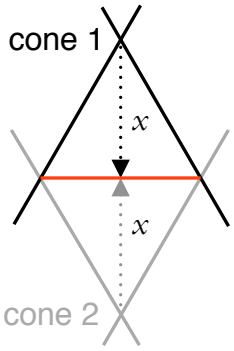
figure 1

For the first placement, it is clear that since both conic surfaces share the same vertex position, no rotation has taken place, and both conic surfaces are identical, therefore every point on each surface is shared. Therefore in case 1, the conic section is the conic surface itself. (see fig. 1)

Case 2 contains two sub-cases: (1) the moved vertex moves up or down along the axis, or (2) lies within the other conic surface but *not* on the axis. I'll first look at the first sub-case.
The radii of the circles that make up a cone (if sliced horizontally, perpendicular to the axis) increase linearly as the slice moves away from the vertex), if two identical, right conic surfaces are observed, then a planar horizontal slice taken a distance *x up* from a vertex will produce a circle identical to if that slice was taken a

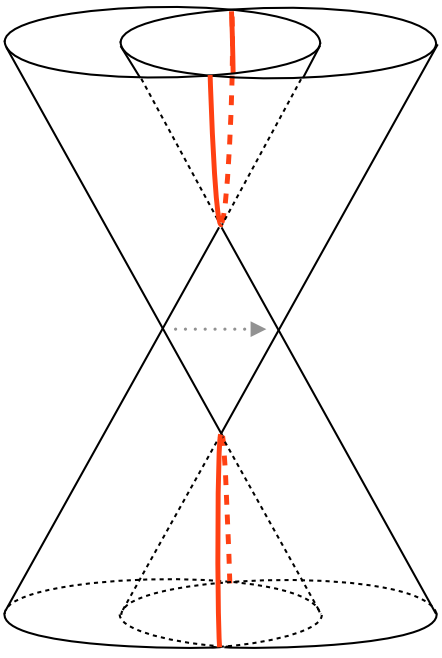


distance *x down* from a vertex. Next, observe the two conics from the side. We see that the generatrices on each side are parallel. Therefore the distance from one vertex to the line connecting the intersections of the generatrices equals the distance from the other vertex to the line. This remains true no matter how far apart the vertices are from each other. Further, this side-view accurately represents matters from any angle. Therefore we can fill in every generatrix on Pascal's projective circle, knowing that it is a true intersection point of both conic surfaces. And since this section is horizontal and perpendicular to the shared axis and lies on one plane, therefore it is a circle. *Quod Erat Demonstrandum.*



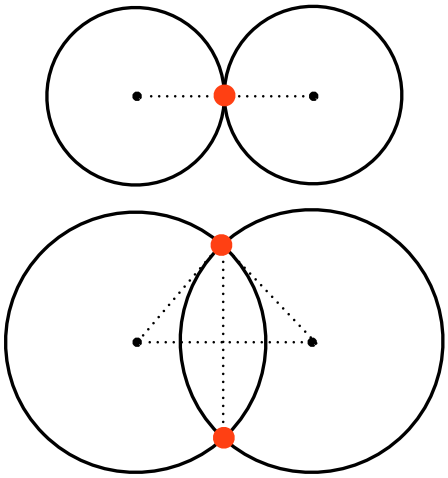
Now for sub-case 2. I hypothesize that the shared section is an ellipse, but I know not how to prove it. I would need to prove that the shared section lies on one plane, and that that plane is oblique to the plane of the axes. I know that the longest diameter of the ellipse would lie in the plane shared by the two axes. Further exploration is needed.

Case 3 has two sub-cases: (1) the moved vertex is moved horizontally, at right angles to the axis, or (2) the moved vertex lies on a different altitude than the original vertex. I'll look at the first sub-case first.

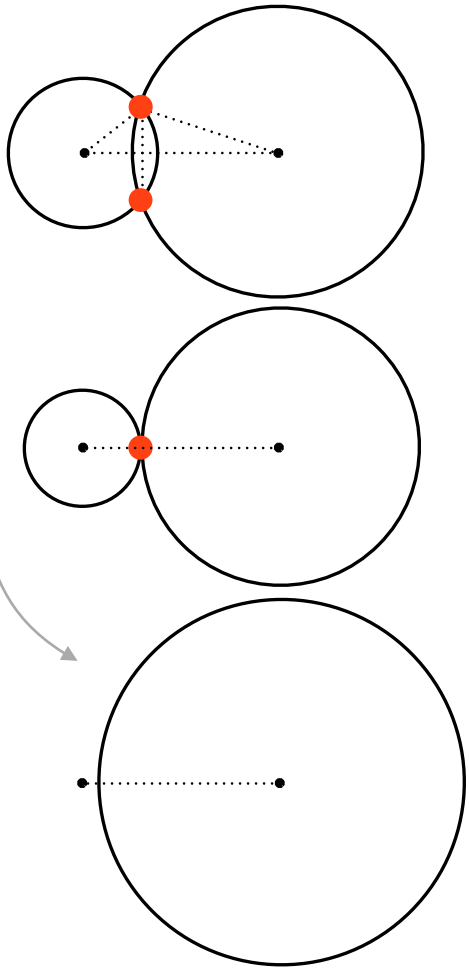


A planar, horizontal slice of a right cone perpendicular to the axis yields a circle as the conic section. With identical conic surfaces, planar horizontal slices taken at the same distance from each vertex will yield circles with equal radii. The two axes are parallel (remain at a fixed distance from each other) but the radii of the identical circles (at any given height above or below the altitude of the vertices) grow linearly in respect to their distance from the vertices. Thus, due to the laws of geometry, no matter how far the two axes are separated, there must be a distance up or

down the axis from the vertices such that the radii of the circles at that altitude equal half the distance between the axes. Therefore at that altitude, the two circles meet but do not cross each other. This point of meeting is exactly halfway between the axes.

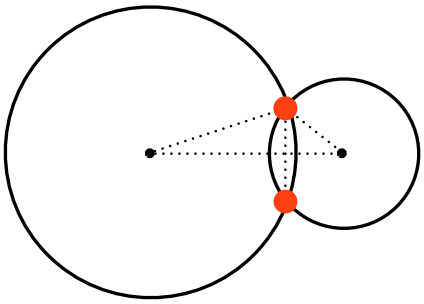


Let us again take a planar, horizontal slice of both conics, but even farther away from the vertices. Looking at this slice from above provides us with a figure similar to this:
It is immediately apparent that the two points of intersection lie in a plane halfway between the axes. Indeed, this construction is merely a form of Euclid's proposition I.10: "*To bisect a given finite straight line*". It is clear, then, that any intersection of identical circles with centers equally separated lies on a line that bisects the line connecting the two centers. Thus the locus of intersections of two identical right conic surfaces with their vertices separated perpendicularly to their axes will lie on a single plane. Therefore this locus is an hyperbola. *Quod Erat Demonstrandum.*



Case 3.2 is more difficult to prove, as we are no longer guaranteed equal radii at a given distance above the vertices. We do know that as the horizontal slice moves closer to both vertices (i.e. the altitude of the slice is not in between the differing altitudes of the vertices) the radii of both circles decreases equally. Once the cutting plane lies on the first vertex it hits (let's call it vertex 1), circle 1 (part of cone 1 which owns vertex 1) is nonexistent and the other (circle 2) still has a radius. At this moment the existing radius for circle 2 ***can never*** equal to the distance between the axes, for if so, then we would be in case 4, and a vertex would be lying on one of the other cone's generatrices. As the cutting plane is moved in between the two vertices, the previously nonexistent circle 1 begins growing at the rate that the circle 2 is shrinking. Following on the previous logic, these two circles are not and can not touch when within these boundaries. Finally the plane hits the other vertex. Now circle 2 is

nonexistent and circle 1 is continuing to grow. Circle 1's radius does not reach vertex 2, as per the logic previously expounded upon. Now the plane passes vertex 2 and recedes from both vertices. Both circles are growing. As the plane continues moving away from both vertices, the circles will grow linearly for infinity. As the growth of these circles is linear, I propose that the intersection points of these linearly growing and shrinking circles lie on a plane oblique to the plane on which the axes lie. Therefore I propose that this locus of intersection points is an hyperbola.



Case 4 is probably the easiest to prove. As one vertex is slid down a generatrix, every generatrix in one conic surface remains parallel to its cousin in the other conic surface, as only translation is taking place. For example: if a line at a given angle and orientation in space is translated to a different place, but kept at the same angle and orientation, it remains parallel to the original line. This is happening on a mass scale in case 4. However, as the vertex is moving down a specific generatrix, that generatrix (being infinitely long) is not moving from itself, but every other generatrix is, so the only shared section is that single generatrix.

Quod Erat Demonstrandum.

